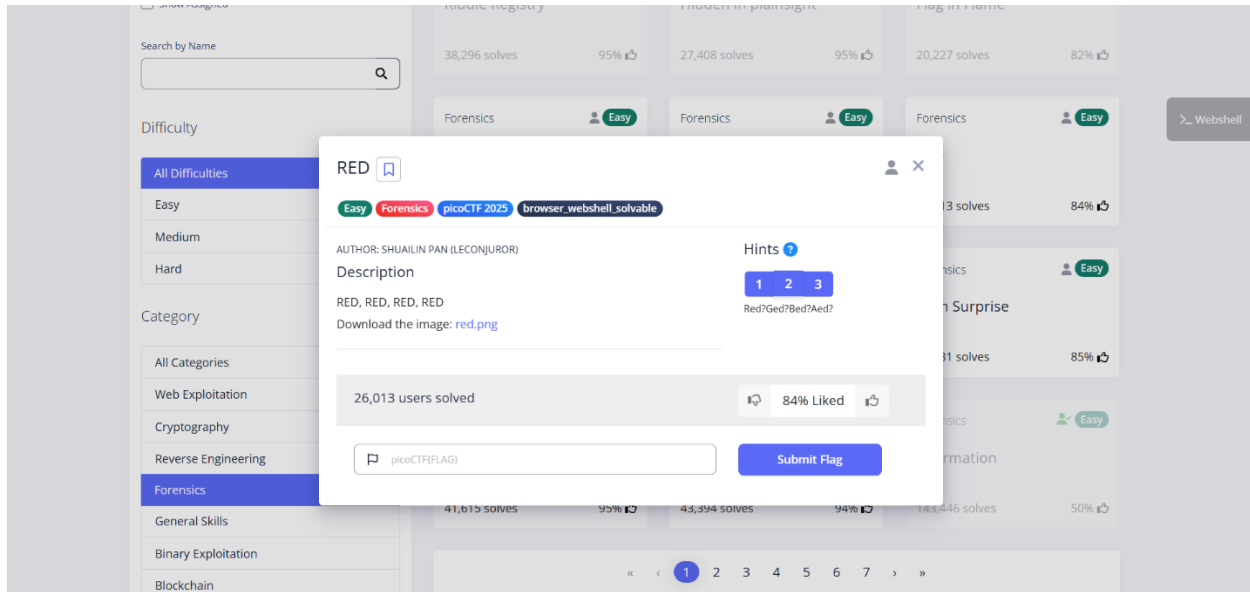




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RED



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Challenge: RED

Category: Forensics

Date: 03/26/26



I. Objective

The objective of this challenge is to analyze an image file and uncover hidden information using **metadata analysis and steganography techniques**.

The goal is to extract hidden data from the image and decode it to **retrieve the flag**.

II. Background

In digital forensics, investigators examine files not just by what is visible, but also by what is **hidden inside them**.

Images can hide information in two main ways:

1. **Metadata** – extra information stored inside the file
2. **Steganography** – hiding data inside the image itself (like secret messages)

In this challenge:

- A clue was hidden in the **metadata**
- The actual message was hidden using **LSB (Least Significant Bit) steganography**
- The extracted data was encoded in **Base64**

III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Working environment
- **exiftool** - Used to extract metadata from the image
- **zsteg** - Used to extract hidden data using steganography

IV. Methodology

1. Launched **Kali Linux** using Oracle VirtualBox



2. Downloaded the file using:

wget <link_address>

```
(kali@kali)-[~/Downloads]
$ wget https://challenge-files.picoctf.net/c_verbal_sleep/831307718b34193b2
--2026-03-26 10:43:21-- https://challenge-files.picoctf.net/c_verbal_sleep/8
Resolving challenge-files.picoctf.net (challenge-files.picoctf.net) ... 3.165.
Connecting to challenge-files.picoctf.net (challenge-files.picoctf.net)|3.165
HTTP request sent, awaiting response ... 200 OK
Length: 796 [application/octet-stream]
Saving to: 'red.png'

red.png                                                    100%[=====]
2026-03-26 10:43:22 (36.1 MB/s) - 'red.png' saved [796/796]
```

3. Extracted metadata using:

exiftool <filename>

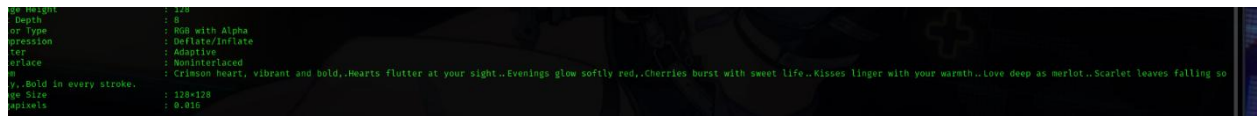
```
(kali@kali)-[~/Downloads]
$ exiftool red.png
ExifTool Version Number      : 13.25
File Name                    : red.png
Directory                    : .
File Size                     : 796 bytes
File Modification Date/Time   : 2025:03:05 22:34:15-05:00
File Access Date/Time        : 2026:03:26 10:43:22-04:00
File Inode Change Date/Time   : 2026:03:26 10:43:22-04:00
File Permissions              : -rw-rw-r--
File Type                     : PNG
File Type Extension          : png
MIME Type                     : image/png
Image Width                   : 128
Image Height                  : 128
Bit Depth                     : 8
Color Type                    : RGB with Alpha
Compression                   : Deflate/Inflate
Filter                        : Adaptive
Interlace                     : Noninterlaced
Poem                          : Crimson heart, vibrant and bold,.Hearts flu
ftly,.Bold in every stroke.
Image Size                    : 128x128
Megapixels                    : 0.016
```

4. Observed a poem in the metadata and noticed that uppercase letters spelled:

CHECK LSB

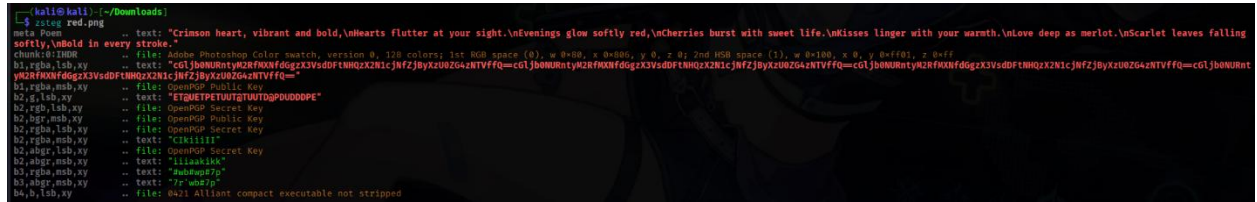


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5. Used zsteg to analyze hidden data:

zsteg <filename>

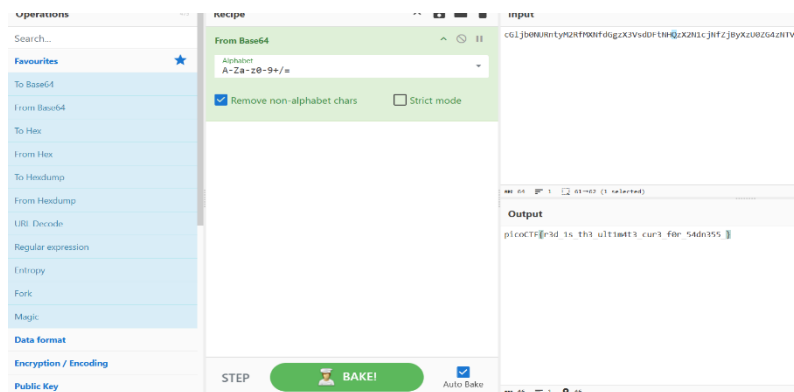


6. Found a hidden encoded message



7. Copied the extracted string

8. Used CyberChef to decode using Base64



9. Successfully retrieved the flag.

V. Result



picoCTF{r3d_1s_th3_ult1m4t3_cur3_f0r_54dn355_}



VI. Explanation

What is Metadata?

Metadata is:

“Information about a file that you can’t normally see”

Example:

A picture may secretly store:

- who created it
- when it was taken
- hidden messages

Think of it like:

A secret note hidden inside a picture

What is exiftool?

exiftool is a tool that:

Shows hidden information inside files

Command:

exiftool image.png

It reveals things like:

- text



- comments
- hidden clues

What is LSB? (Very Simple)

LSB = Least Significant Bit

Sounds complex? but here's the easy version:

Think of a pixel like a number:

10101010

The **last digit (rightmost)** is the LSB

What do hackers do?

They:

Hide secret data in those tiny last bits

Because:

- It doesn't change how the image looks
- But it stores hidden data

What is zsteg?

zsteg is a tool that:

Finds hidden messages inside images using LSB

Command:

zsteg image.png

It scans the image and tries to extract hidden data.



If zsteg is NOT installed

You can install it using:

sudo apt install zsteg

What is Base64?

After extracting the hidden message, it looked like random text.

That's because it was encoded in **Base64**

VII. Conclusion

This challenge demonstrates how data can be hidden in multiple layers within a file.

By analyzing metadata and applying steganography techniques, the hidden message was successfully extracted and decoded.

This reinforces an important concept in digital forensics:

Not all data is visible — sometimes the most important information is hidden beneath the surface.

— The Analyst: Hyposelenia